Sijia (Lucia) Liu

Ph.D. Candidate in Chemistry, University of California, Irvine

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Education

University of California, Irvine — *Irvine, CA*

Ph.D. in Chemistry — *Sept 2022 – June 2027 (expected)*

Advisors: Prof. Sergey A. Nizkorodov and Prof. Celia L. Faiola

M.S. in Chemistry — *Sept 2022 – June 2024*

B.S. in Chemistry, Cum Laude · *Minor in Psychology* — *Sept 2017 – Dec 2021*

Awards and Fellowships

Best Poster Award, International Aerosol Conference (IAC) — *Sept 2026*

Xi'an, China

International Summer School on Air-Climate-Health, Selected Participant — *Aug 2026*

Peking University, Beijing, China

DOE ARM Summer School: Big Open Data Science, Selected Participant — *May 2026*

National Weather Center, Norman, OK

F. Sherwood Rowland Graduate Research Fellowship — *2025 – 2026*

University of California, Irvine

AAAR Conference Travel Award — *Oct 2023 · Oct 2025*

UCI Chemistry Travel Award — *Oct 2023 · Oct 2025*

Teaching Program Contribution Award — *May 2023*

Department of Chemistry, University of California, Irvine

Research Experience

PhD Intern — *2024 – 2025*

Environmental Molecular Sciences Laboratory, Pacific Northwest National Laboratory — *Richland, WA*

Advisors: Dr. Alla Zelenyuk and Dr. Swarup China

- **VCP-Induced Aerosol Formation Indoors.** Designed indoor experiments simulating volatile chemical product use under realistic indoor ozone levels, and characterized rapid SOA formation, new particle formation, particle volatility, and the effects of ventilation using miniSPLAT, SEM, and nanoDESI-HRMS.
- **TD-GC-QToF-MS Method Development.** Led installation and calibration of a TD-GC-QToF-MS and sorbent tube conditioner, and developed workflows to quantify VOCs in combustion, chamber, and field samples across multiple ARM campaigns.
- **Vertical VOC Profiling with TRAVIS.** Co-led TD-GC-QToF-MS analysis of TRAVIS sorbent tube samples to evaluate system performance for altitude-resolved VOC profiling.
- **Temperature- and Size-Resolved Characterization of Biomass Burning Aerosol.** Led an interdisciplinary study of pine needle combustion aerosol, resolving size- and temperature-dependent particle properties with nanoDESI-HRMS, SEM, and miniSPLAT, in collaboration with Dr. Manish Shrivastava.
- **Sesquiterpene Ozonolysis SOA.** Designed chamber experiments at varying precursor concentrations and characterized SOA morphology, volatility, and chemical composition using miniSPLAT and nanoDESI-HRMS.

- **Above- and Below-Cloud Aerosol at ARM Sites.** Led molecular and particle-level analysis of ArcticShark UAS samples from DOE ARM sites to evaluate vertical variability and aerosol–cloud interactions using nanoDESI-HRMS, SEM, STXM, and meteorological data.
- **SOA Formation over the Southeast United States.** Processed terpene SOA volatility basis set parameters for WRF-Chem simulations centered on the ARM Bankhead National Forest site, in collaboration with Dr. Manish Shrivastava.

Graduate Student Researcher *Sept 2022 – Present*
University of California, Irvine *Irvine, CA*
Advisors: Prof. Sergey A. Nizkorodov and Prof. Celia L. Faiola

- **SOA from Photooxidation of BVOC Mixtures.** Led chamber studies of VOC oxidation, chemical cross-reactions, and SOA formation in biogenic VOC mixtures, tracking precursor and product evolution with TD-GC-MS and PTR-ToF-MS, characterizing particle composition with UPLC-HRMS, and interpreting mechanisms with GECKO-A modeling, in collaboration with Drs. Tommaso Galeazzo, Richard Valorso, and Manabu Shiraiwa.
- **SOA from AVOC and BVOC Mixtures.** Mentored a Beckman Scholar in chamber experiments and HRMS analysis comparing SOA yields and composition from anthropogenic and biogenic VOC mixtures.
- **Particle-Phase Mixture Effects in SOA.** Investigated interactions between SOA from different VOC precursors under neutral and mildly acidic conditions using PTR-ToF-MS and UPLC-HRMS.
- **Properties of Cyclic and Acyclic BVOC SOA.** Characterized physical and chemical properties of monoterpene- and sesquiterpene-derived SOA using single-particle mass spectrometry and HRMS, in collaboration with Dr. Alla Zelenyuk.

Junior Specialist *Jan 2022 – Sept 2022*
Undergraduate Student Researcher *Aug 2021 – Dec 2021*
University of California, Irvine *Irvine, CA*
Advisor: Prof. Sergey A. Nizkorodov

- **SOA Aging under Acidic Conditions.** Investigated the compositional evolution of SOA under acidic conditions using UV-Vis spectroscopy and HPLC-PDA-HRMS.
- **Impact of Exceptional Events on Air Quality.** Analyzed EPA data in R to evaluate state-reported exceptional event claims and their implications for air quality policy.

Peer-Reviewed Publications

1. **Sijia Liu**, Claire Moffet, Gregory W. Vandergrift, Zezhen Cheng, Swarup China, Manish Shrivastava, Sergey A. Nizkorodov, Alla Zelenyuk, and Celia L. Faiola. *Secondary Organic Aerosol from OH Oxidation of Acyclic Terpenes Is More Viscous and Less Volatile than That of Their Cyclic Analogs.* ACS ES&T Air **2026**, 3(1), 83–94.
2. Gregory W. Vandergrift, **Sijia Liu**, Andrey V. Liyu, Zezhen Cheng, Casey Longbottom, Damao Zhang, Darielle N. Dexheimer, and Swarup China. *Demonstration of an Automated System for Vertical Profiles of Volatile Organic Compounds.* ACS Earth Space Chem. **2025**, 9(11), 2597–2606.
3. **Sijia Liu**, Tommaso Galeazzo, Richard Valorso, Manabu Shiraiwa, Celia L. Faiola, and Sergey A. Nizkorodov. *Secondary Organic Aerosol from OH-Initiated Oxidation of Mixtures of D-Limonene and Myrcene.* Environ. Sci. Technol. **2024**, 58(30), 13391–13401.
4. Cynthia Wong, **Sijia Liu**, and Sergey A. Nizkorodov. *Highly Acidic Conditions Drastically Alter the Chemical Composition and Absorption Coefficient of α-Pinene Secondary Organic Aerosol.* ACS Earth Space Chem. **2022**, 6(12), 2983–2994.

Manuscripts Submitted or Under Review

1. **Sijia Liu**, Gregory W. Vandergrift, Zezhen Cheng, Nurun Nahar Lata, Ashfiqur Rahman, Valentina Sola, Zhenli Lai, Xena Mansoura, Nicole Riemer, Daniel A. Knopf, Beat Schmid, Matthew A. Marcus, Damao Zhang, Alla Zelenyuk, Fan Mei, Celia L. Faiola, Sergey A. Nizkorodov, and Swarup China. *Vertically Resolved Measurements Reveal a Chemically Distinct Aerosol Layer Adjacent to Clouds.* Submitted to Environ. Sci. Technol., **2026**.
2. **Sijia Liu**, Avisa Charmchi, Celia L. Faiola, and Sergey A. Nizkorodov. *Effects of Cross-Reactions on Chemical Composition and Properties of Particulate Matter Produced by Oxidation of a Mixture of Anthropogenic and Biogenic Volatile Organic Compounds.* Submitted to ACS ES&T Air, **2026** (ea-2026-00357p).

Manuscripts in Preparation

1. **Sijia Liu**, Amna Ijaz, Zezhen Cheng, Sergey A. Nizkorodov, Celia L. Faiola, Manish Shrivastava, and Alla Zelenyuk. *Rapid Secondary Organic Aerosol Formation from Volatile Chemical Product Sprays under Realistic Indoor Ozone Levels.* To be submitted.
2. **Sijia Liu**, Amna Ijaz, Sergey A. Nizkorodov, Celia L. Faiola, Manabu Shiraiwa, and Alla Zelenyuk. *Precursor Concentration and Molecular Structure Control the Viscosity and Volatility of Secondary Organic Aerosol Derived from Sesquiterpene Ozonolysis.* In preparation.

Software and Data Products

1. Maggie Powell, Yijia Sun, Paul Pappu, **Sijia Liu**, Emmanuel Kipchirchir, Luojie Dong, Jing Li, Luke Heim, Sining Niu, William Gustafson, Hsi-Yen Ma, and Joseph R. O'Brien. *2026 ARM Summer School Project. Model Meets Reality (MMR): A Cross-Comparison of DP-SCREAM and WRF Against ARM Observations.* Zenodo, **2026**. DOI: 10.5281/zenodo.21113437.

Conference and Invited Presentations

Nanjing–Helsinki Institute, Invited Seminar *Suzhou, China · Sept 2026*

Invited Seminar. Secondary Organic Aerosol from Structurally Distinct Volatile Organic Compounds: From Single Precursors to Mixtures

International Aerosol Conference (IAC) *Xi'an, China · Sept 2026*

Poster. Secondary Organic Aerosol from OH Oxidation of Acyclic Terpenes Is More Viscous and Less Volatile than That of Their Cyclic Analogs **(Best Poster Award)**

Pacifichem Conference *Honolulu, HI · Dec 2025*

Outdoor/Indoor Atmospheric Chemistry Symposium

Oral. Size-Resolved Chemical Composition and Physical Properties of Biomass Burning Aerosol Produced at Different Burning Temperatures

Poster. Precursor Concentration Modulates Evaporation Kinetics and Chemical Composition in Acyclic and Cyclic Sesquiterpene-Derived SOA via Ozonolysis

American Association for Aerosol Research (AAAR) Annual Conference *2022 – 2025*

2025 *Buffalo, NY*

Oral. Precursor Concentration and Molecular Structure Influence the Viscosity and Volatility of SOA Derived from Sesquiterpene Ozonolysis

Poster. Contrasting Above- and Below-Cloud Aerosol Composition and Properties from ArcticShark UAS at Bankhead National Forest

2024 *Albuquerque, NM*

Oral. A Comparative Study on the Evaporation Kinetics and Physical Properties of Acyclic and Cyclic Terpene SOA

Poster. Investigating Particle-Phase Chemistry between Products of Oxidation of α-Pinene and Phenol

2023 *Portland, OR*

Oral. Acyclic Monoterpenes Do Not Suppress SOA from Cyclic Monoterpenes

x **Invited Talk.** Secondary Organic Aerosol from Stressed Emissions (PNNL EMSL Info Session)

2022 *Raleigh, NC*

Poster. Effect of Cross-Reactions in Monoterpene Mixtures on SOA Formation

EMSL User Meeting *Richland, WA · 2024*

Oral. A Comparative Study on the Evaporation Kinetics and Physical Properties of Acyclic and Cyclic Terpene SOA

Informal Gathering on Atmospheric Science and Photochemistry (IGASP) *Los Angeles, CA · 2023*

Poster. SOA from Biogenic VOC Mixtures

Technical Skills

VOC Measurement

PTR-ToF-MS, TD-GC-QToF-MS (installation, calibration, and quantification workflows), TD-GC-MS, sorbent tube sampling and conditioning

High-Resolution Mass Spectrometry and Chromatography

UPLC-PDA-HRMS (Orbitrap), ESI-HRMS, nanoDESI-HRMS, SPLAT, miniSPLAT, AMS

Aerosol and Atmospheric Instrumentation

Environmental chamber, indoor VCP simulation experiments, PAM-OFR, continuous combustion chamber, SMPS, MOUDI, SEM, STXM, UV-Vis

Modeling, Programming, and Data Analysis

GECKO-A, WRF-Chem VBS parameterization, Python, R, MATLAB, Mathematica, LabVIEW, Igor Pro, OriginPro, Git

Instrument and Vendor Software

Thermo Xcalibur, FreeStyle, Agilent MassHunter, PTR-MS Viewer, MZmine, ChemDraw

Languages

Chinese (native), English (fluent), Japanese (intermediate)

Mentorship of Undergraduate Researchers

Avisa Charmchi, UCI Beckman Scholar *June 2024 – July 2025*

Mentored chamber experiments and HRMS analysis, resulting in a co-authored submitted manuscript.

Zhonghe Qin, UCI UROP Fellowship Awardee *May 2023 – May 2024*

Teaching Experience

Teaching Assistant, University of California, Irvine *Irvine, CA*

CHEM 1A · General Chemistry I *Winter 2024*

CHEM 153 · Advanced Analytical Chemistry Laboratory *Spring 2023*

CHEM 1LE · General Chemistry Laboratory *Winter 2023*

CHEM 1X · *General Chemistry Extra* *Fall 2022*

Leadership and Service

AAAR Student Media Ambassador *2026*

Represent the American Association for Aerosol Research on social media, communicating aerosol science and conference highlights to broad audiences.

Certified UHPLC-PDA-HRMS Instrument Operator, AirUCI *Oct 2022 – Present*

Maintain and calibrate Orbitrap HRMS instrumentation, customize analytical methods for users, and coordinate quarterly meetings on system status and training.

Professional Memberships

American Association for Aerosol Research (AAAR) *2022 – Present*

American Chemical Society (ACS) *2022 – Present*